

Enterprise configuration management — controlling integration complexity

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Greater use of flexible system architectures, and changes in product acquisition policy, are just two elements of delivery strategy that have led to a significant increase in systems integration scale and complexity. This has produced the need for a new approach to the way BT manages and controls system whole-life change. This paper describes a major BT initiative that will provide a configuration management environment capable of catering for many different viewpoints — from strategic planning through to operational deployment and eventual system withdrawal.

1. Introduction

All organisations must adapt to change if they are to survive in to-day's harsh, commercial, climate but this is even truer for those large companies whose very existence is dependent on modern technology and innovation. In particular, this applies to national and international telecommunications companies. Two years ago the BT systems encyclopaedia [1] contained details of approximately 1350 operational support systems (OSS). One year later the number had risen to 1750 and, today, there are well over 2000. At this rate the company's portfolio of systems will double in three years with all the attendant maintenance and support costs; not to mention the growth in inter-system dependencies (e.g. system interfaces and project management). There are about 6000 people in the Systems Engineering department of BT's Networks and Systems (N&S) Division, pouring out tens of thousands of lines of code daily. It is a potential systems integration nightmare and enough to frighten the faint-hearted.

How can this growth in system complexity be managed and controlled? How can the growth rate be slowed without compromising the ability to meet market and regulatory demands? How can the enormous investment in people, time, and money be used ever more effectively, to the benefit of customer and company alike?

The solution lies in the domain of configuration management (CM). The answer is not simply a matter of acquiring a robust CM tool capable of dealing with many millions of configuration items such as source code files, specification and design documents, deployed configurations, and inventories. Any workable solution must cater for the diversity of the enterprise and its suppliers, together with the wide range of acquisition, maintenance, and support processes, both internal and external to the

company. This is the difference between enterprise configuration management (ECM) and traditional, project or deployment, CM.

BT's enterprise CM solution is the configuration management environment (CME) [2]. This provides a whole-N&S, birth-to-death, technological and process environment capable of catering for many different operational and managerial viewpoints — from evolution planning through to operational deployment and eventual system withdrawal.

This paper discusses the need for ECM within BT and the future of enterprise CM within the company.

2. The need for enterprise CM

The way BT delivers and supports its network and information systems has changed over the past few years to meet external and internal demands for new and better solutions, delivered in shorter time-scales, and to ever higher quality standards. Greater use of flexible system architectures, and changes in product acquisition policy, are just two elements of delivery strategy that have led to a significant increase in systems integration.

Integration and deployment, which, a relatively short time ago, were constrained to small functional modules is taking place at the service platform level (see section 4). All of this has produced the need for a new approach to the way system whole-life change is managed and controlled.

Although the CM difficulties associated with large-scale telecommunications systems integration and deployments are recognised within the IT industry, few organisations

have even attempted to resolve the problems on the scale required by BT [3]. The aerospace and vehicle manufacturing industries have had most success when it comes to hard components such as wheels, doors, nuts, and bolts. There is a lot to be learnt from these sources but CM of a software-critical IT solution is significantly more difficult by virtue of their flexibility, complexity and the soft nature of the components.

Large telecommunications companies that do not address these problems suffer serious gaps in their ability to capture and control significant portions of their networks and systems infrastructure knowledge base. This in turn leads to weaker solution designs, more costly implementations, increased operational costs, and unreliable evolution plans. On the other hand, the telco that solves the enterprise CM problem first will have a competitive advantage. There are significant productivity gains in production and deployment, and lower failure costs through better design, diagnostic, and repair capabilities.

3. Control and ECM

3.1 What is control?

The word 'control' often has an emotive connotation and interpreted by some people to suggest direction or command by giving orders. Others may be suspicious of control systems and see them as an indication of authoritarian management or as a means of exerting pressure or maintaining discipline. There is far more to control than simply a means of restricting behaviour or the exercise of authority over others. Enterprise CM provides a general type of control applied to both individual behaviour and organisational performance. Its design and implementation is constructive and rewarding. In this way it caters for positive, as well as negative, effects.

Reliability, order, and stability are words often associated with the need for control. Programme, project, and integration managers will ask, among other things, 'What milestone or scope targets do I have to meet?' 'What versions of each component do I need to integrate if I am to meet system scope objectives?' 'How and when do I join them together?' 'How well am I doing so far?' This is asking for control. Operational and production managers want to know how well they are performing and, in turn, this places emphasis on the exchange of information, feedback, and the comparison of actual results against planned targets. At the same time, the company must exercise control over the behaviour and actions of people in order to ensure a satisfactory level of organisational performance in a rapidly changing, highly technical environment.

An important factor in the evolution of BT's enterprise CM solution, CME, over the past eighteen months or so, has

been the concern that most people are ambivalent to control. While they may not wish to have control applied to their own performance they recognise the usefulness of and need for some sort of control over other systems or organisations on which they are dependent. Lawlor [4] gives three reasons why people and organisations need control systems:

- to give feedback about task performance,
- to provide some degree of structure of tasks, definition of how tasks are to be carried out, and how performance will be measured,
- to evaluate performance, especially in reward systems, such as pay.

The whole purpose of control, and hence enterprise CM, is to improve the performance of the individual and the organisation, in this case BT (see Fig 1).

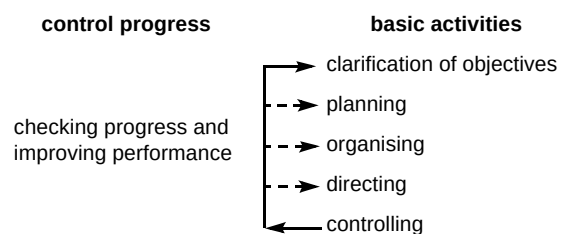


Fig 1 A typical control process.

3.2 Approaches to control

There have been many studies since the beginning of this century concerned with the achievement of optimal control — from the classical, technological approach to the contingency, depends-on-the-situation approach.

The classical approach [5] places emphasis on the planning of work, the technical requirements of the organisation, principles of management, and the assumption of rational and logical behaviour. Emphasis is on the hierarchical nature of management and formal organisational relationships.

The contingency approach [6] highlights the complex nature of systems and organisations. It takes the view that there is no one, best, universal form of control structure; there are a large number of variables that influence behaviour and performance so it's best to wait-and-see. *If* it's working don't fix it, *else* react to those variables that are most important at the time.

Neither of these approaches provides a workable basis for an enterprise CM solution because one is too prescriptive and the other is too reactive. To meet BT's complex systems integration and rapid-response-to-market objectives the environment must be proactive. It must

consider the behaviour of those who exercise control and those who are subject to control.

Between the classical and contingency extremes there is another significant control theory. The open-systems theory better describes the approach adopted as the foundation of ECM. Control in the ECM context relates to both end-results and to the processes by which inputs are transformed to outputs. In ECM, control is the mediator between the input and output processes. It helps to maintain the working of integrated systems and their production, operational, and organisational control.

3.3 Elements of enterprise CM

Enterprise CM, and, in particular, the BT implementation, CME, is not a technological problem. Its success or failure depends mostly on process and behaviour variables. The underlying principles are best described in terms of the five essential elements of a control system (Fig 2) [7], consistent with an open-systems approach:

- planning the desired (e.g. requirements, scope, solutions, delivery, operational support, time-scales, milestones, risk and contingencies, configuration identification),
- establishing performance standards (e.g. deliverables at appropriate life cycle stages, naming conventions, common processes, risk analysis criteria, configuration control),
- proactive monitoring of actual performance (e.g. milestone reminders, automatic progress report generation, flexible disaster avoidance strategies, status accounting),
- comparing actual with planned targets (e.g. maintenance of historical data, progress reporting, priority analysis, baselining, configuration auditing),

- taking corrective action (e.g. adjusting priorities based on effective decision support, rapid redeployment of resources in business-critical scenarios).

If an enterprise CM system is to be effective it must also be 'understandable and economical, related to decision centres, register variations quickly, be selective, remain flexible and point to corrective action' [8].

4. The platform framework

The 'platform framework' is a radical new approach to delivering the networks and systems needed by BT. These networks and systems are the engineering solutions to product requirements defined by BT's customer-facing divisions. They have to evolve in the time-scales dictated by today's market-place rather than the time-scales traditionally associated with the deployment of telecommunications infrastructure.

The platform framework provides a way of simplifying and managing the complex inter-relationships between products, network technologies and operational support systems. This will reduce the time to market, cut costs and provide clear traceability from product requirements through to network and system capabilities.

A service platform is the integrated set of components including network equipment, operational support systems and operational processes, which, taken together, provide an environment in which sellable BT services can be delivered [9].

4.1 The challenges

Today's communications environment presents a number of major challenges to BT.

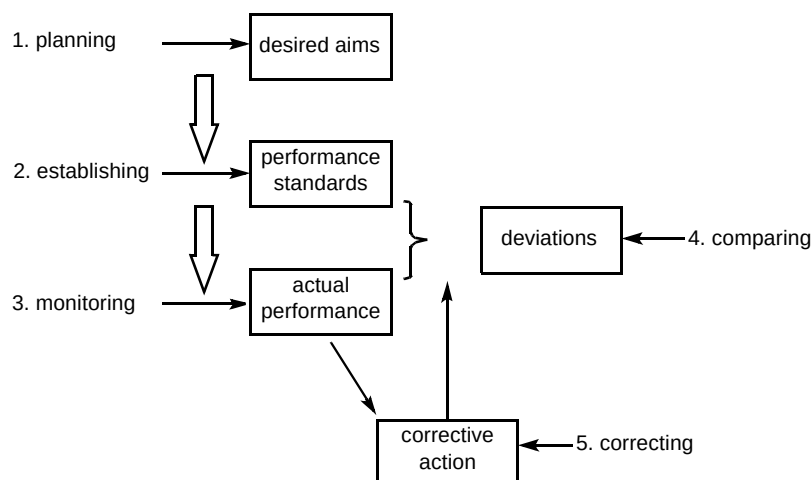


Fig 2 Elements of a control system.

Reducing time to market by an order of magnitude

Markets evolve far more quickly than telcos can deliver engineering solutions to their customers. Products are characterised by:

- short life,
- high current account investment in R&D,
- short time to market;

whereas infrastructure is characterised by:

- long life,
- large capital investment,
- long deployment times.

Reducing the expenditure on infrastructure

Tactical network and management solutions are used to get products launched on time. This makes sound commercial sense in the short term but creates problems in the longer term. If the tactical solutions remain as stove-pipes (stand-alone solutions) the consequence is too many platforms, each using up skilled resources to design, build and run it.

The result is high unit costs (per line or per call) because the stove-pipe solution has to recover all the procurement and operational costs from a very limited set of products. This makes large communications companies such as BT vulnerable to fresh-start niche operators with low overheads. They are prevented from exploiting the advantages of a coherent product portfolio and wide geographical presence.

Although overlays will always be necessary, especially in the pilot/nursery stage of product development, the aim must be to maximise the number of products and services delivered on each of a (relatively) small number of platforms.

Delivering complete solutions

Network systems and operational support systems (OSSs) tend to be deployed to the operational units as individual items rather than complete solution sets. The integration teams do their best to bring all the piece-parts together, but an integrated approach is needed throughout the life cycle — evolution planning, design, programme management, etc. Companies must think in terms of complete solutions — an integrated set of networks and systems, OSS and processes which are upgraded in a controlled way.

Understanding true cost structures

Where accounting separation is in force companies must have a clear and auditable relationship between the products

delivered by individual trading units and the component technologies. This is not a simple mapping, since many products make use of other products that may in turn rely on yet other products. The successful model will satisfy the accounting needs of the business. In BT's case, it is not just a question of meeting the demands of the regulator — the company also needs to generate accurate cost information for complex investment decisions.

Reducing programme delivery risks

Late delivery of large-scale, integrated solutions is a perpetual problem within the telecommunications industry and a major factor is the lack of company-wide configuration management systems. Computer systems engineering enterprises need to apply configuration management with the professionalism and discipline of leading engineering and manufacturing companies.

4.2 Platform layering

Figure 3 shows the relationship between products and services and networks and systems, without the platform framework. The complexity is unmanageable.

Figure 4 shows a similar picture but this time the service platform concept has been introduced. It is relatively easy to map products to individual service platforms. However many platforms rely on other platforms to deliver service. Each platform is therefore made up from a common set of core components (network systems and OSS), plus 'sub-contracted' products from other platforms or sub-platforms where necessary.

The layering approach is powerful in a number of ways:

- it introduces complexity breaks at practical and implementable boundaries,
- it provides two-way traceability between the end products through the various platforms to individual system developments and procurements,
- costs can be tracked along the value chain so that investment decisions can be based on a clear understanding of the profitability of products and solutions,
- it provides a model for cost accounting that more accurately reflects the apportionment of costs to revenues.

Approximately one hundred national and international service platforms have been identified within BT's global enterprise to date.

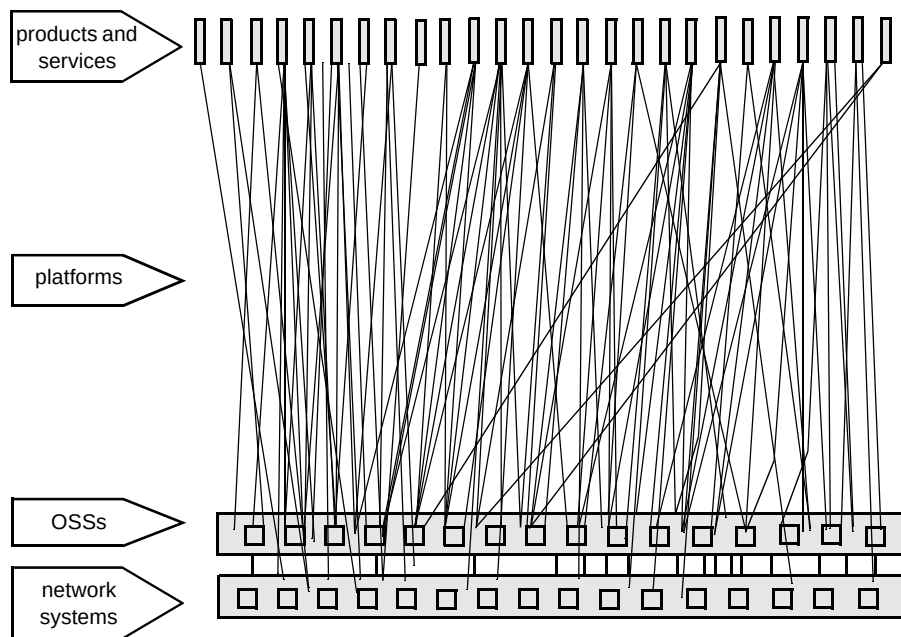


Fig 3 Relationships without platforms.

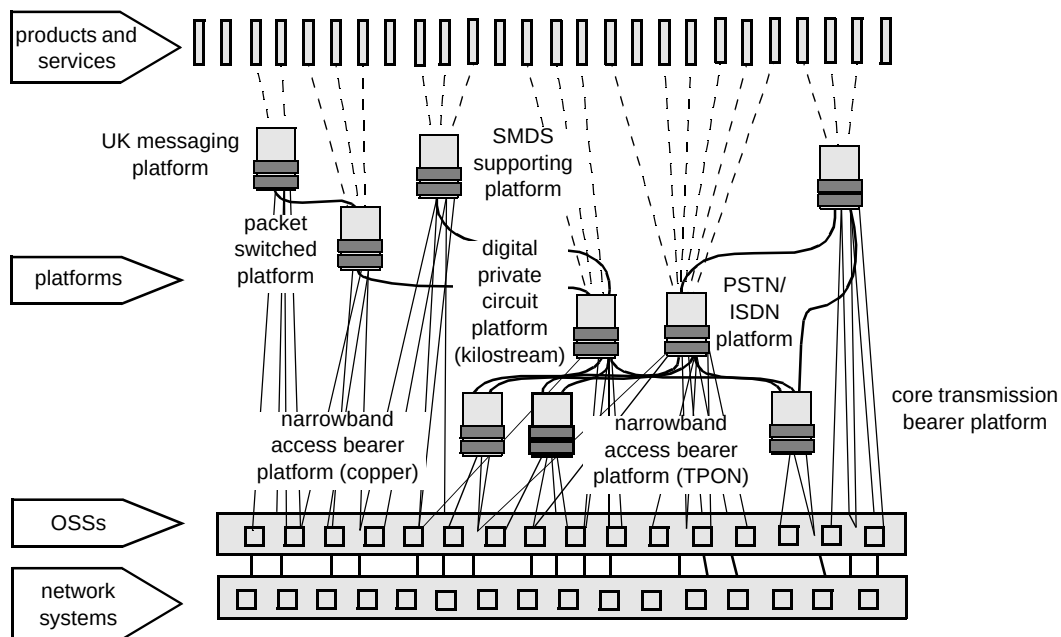


Fig 4 Relationships with platforms and layering.

4.3 The 'flowerpot' diagram

Products and services are the lifeblood of most commercial enterprises and in telecommunications companies these products and services rely heavily on their supporting networks and operational support systems. It is no longer viable to engineer solutions from scratch for every new demand. Stiff competition, ever-decreasing time-to-market and sophisticated requirements have seen to that. A rich variety of new products and services must be produced from fewer physical components. This can only be achieved using flexible designs and rapid reconfiguration

of underlying network and operational support system components. This leads to the 'flowerpot' diagram shown in Fig 5.

Unfortunately, this flowerpot arrangement is not common in large telecommunications enterprises. Often, the flowerpot is then up-ended. As pointed out in the introduction to this paper, BT's operational support system portfolio increases at approximately 20% each year. The reason for this, as with many of the other difficulties mentioned previously, is the complex, interactive nature of modern IT solutions. In the past, marketable products and

services could be engineered within a single production environment and frequently within a single system. Today, a single, major product or service is often distributed over many large, integrated systems.

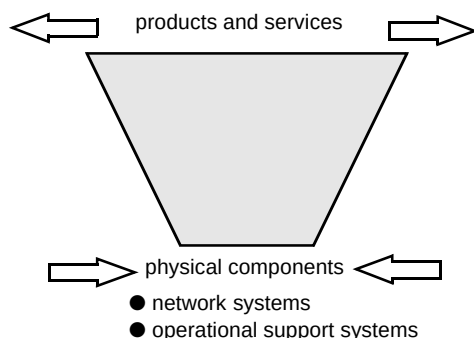


Fig 5 The 'flowerpot' diagram.

It is the concept of a logical, higher level, component based on business need that provides a context within which the BT enterprise CM system can be made to function effectively. Service platforms and ECM are dependent on each other; service platforms because they are too complex to manage manually, and ECM because it needs a relationship between physical components and business requirements.

4.4 Service platform CM

Figure 6 shows a typical service platform, the Switched Multi-megabit Data Services Platform (SMDS), and some of its components. Here the configuration problem is viewed from the platform perspective with several versions of the platform cutting across several versions of the underlying components. As the platform evolves, the number and variety of components increase and the ability to control change without affecting overall performance and quality becomes more difficult. It must be remembered that each of the components may be a large, integrated system in its own right (e.g. COSMOSS).

The other side of the story, from the component's perspective, is shown in Fig 7. Here, various versions of the COSMOSS component are being delivered with each version meeting the needs of various platforms, including the SMDS platform.

Which platform has priority when difficulties arise in production of the component? What is the impact on the platforms, and hence the enterprise, of a failure to meet component delivery targets? Who is responsible for delivery at each and every stage of a complex platform integration and deployment? Who authorises changes and when? How can individual life cycle milestones be

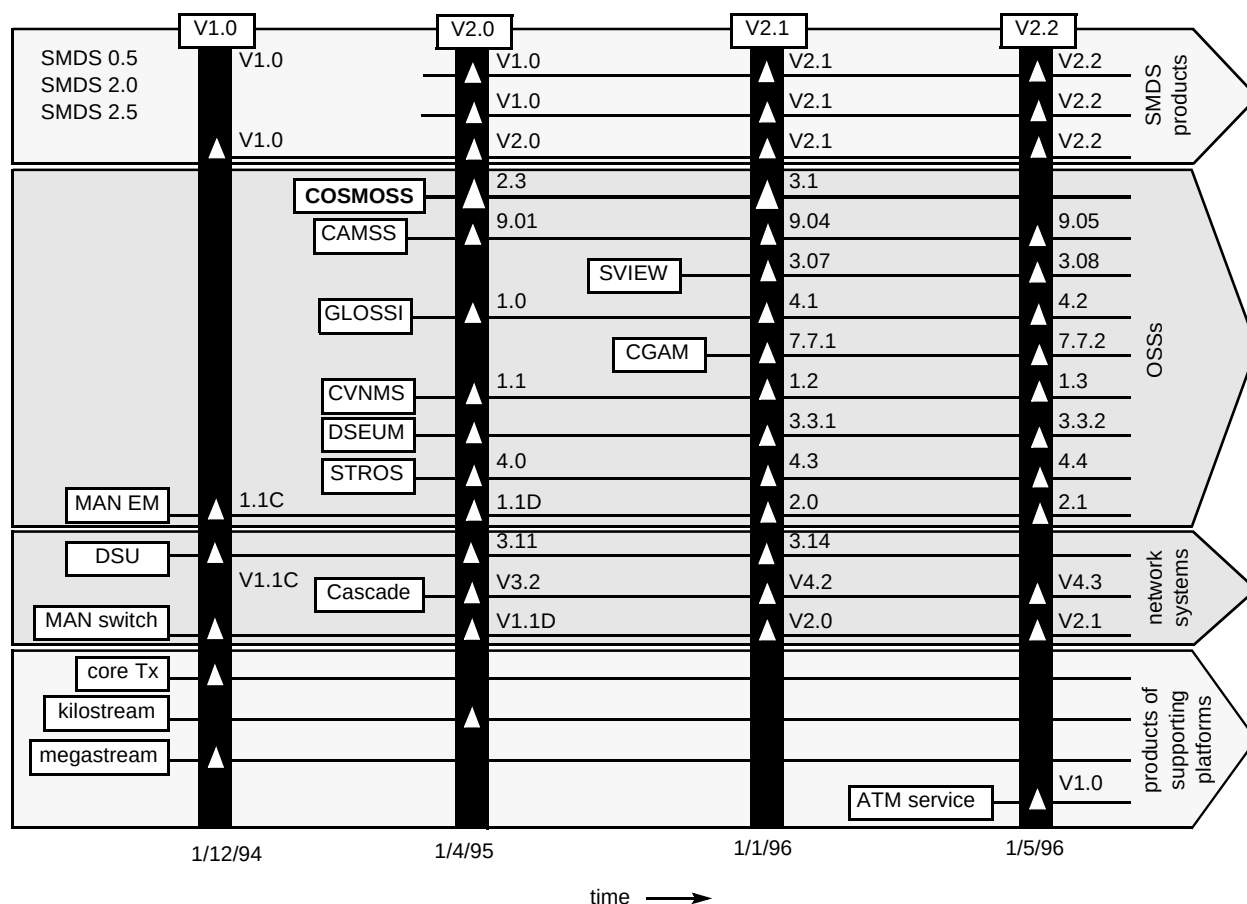


Fig 6 A typical service platform.

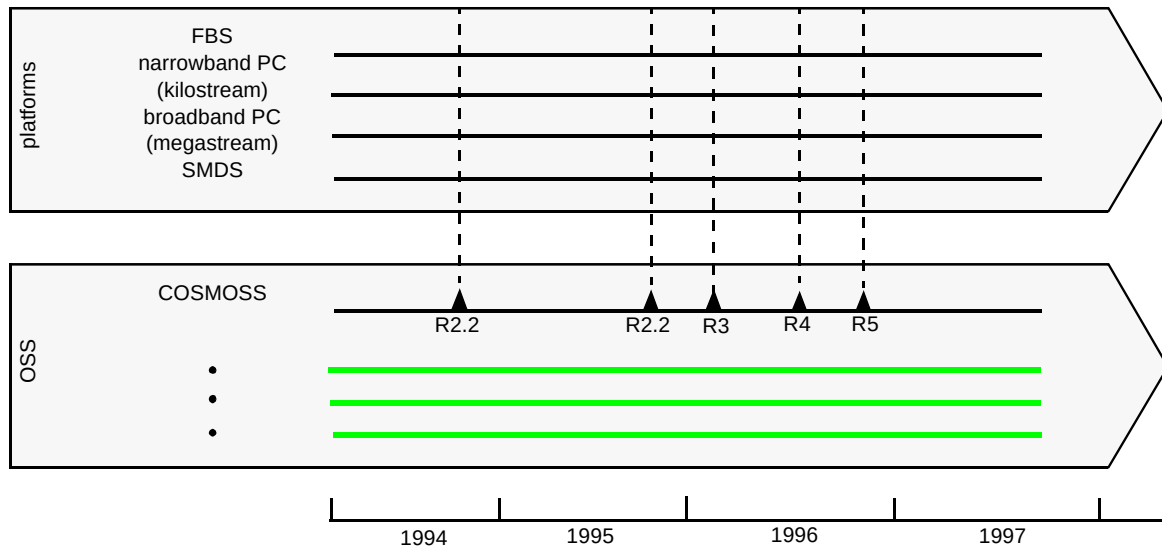


Fig 7 A typical platform component.

controlled? How can change be proactively monitored and controlled?

These are some of the issues to be addressed if the global telecommunications enterprise is to function efficiently. The service platform concept and ECM provide a framework within which to engineer solutions.

5. CM hierarchies

The basic activities of configuration management (identification, control, status accounting, and audit) have been carried out by engineering teams for many years. However, the target for control has mainly been physical items (e.g. files, documents, hardware, etc). As computer systems and organisations become more complex and highly integrated the need for similar activities targeted at more logical entities (e.g. service platforms, assemblies of operational sup-

port systems, individual requirements, and capabilities) becomes inevitable. At the same time, both these sets of activities must be co-ordinated, co-operative, and consistent. This leads to a hierarchy of configuration management as shown in Fig 8.

As with systems and their hierarchy of modules and components, there is a higher level hierarchy composed of business solution and service platform components. Traditional, or project and deployment, CM is mostly carried out at the lower levels while the configuration management environment primarily focuses on the higher levels in the CM hierarchy.

Some project CM activities, such as 'build control', are not performed in the CME and the nature of other activities, such as 'problem management', is quite different. For example, problem management at the lower levels may be

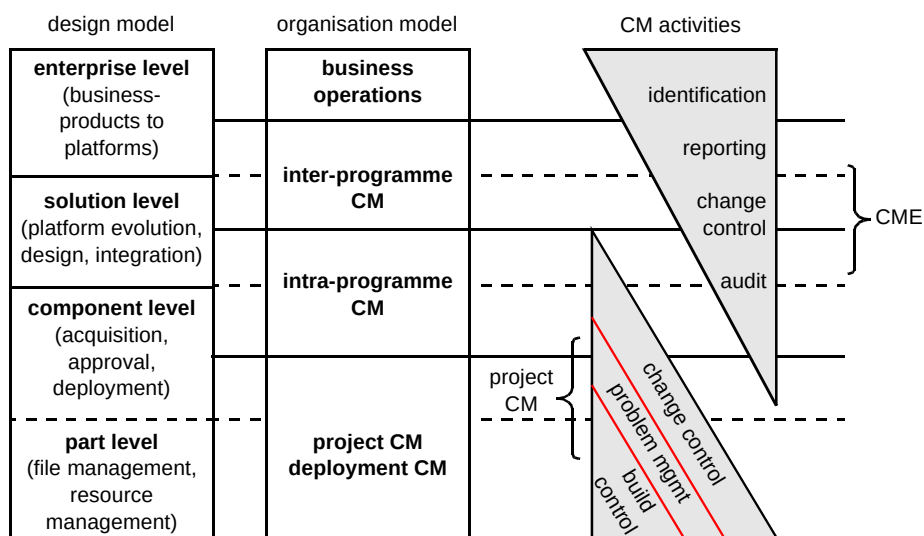


Fig 8 The CM hierarchy.

concerned mainly with module or component bug reports while the same activity in CME will focus on systems integration and configuration problems. There is a significant overlap between project CM and CME and the two together provide the BT enterprise CM system.

6. ECM — problem definition

By the end of 1995 it was clear within BT's Design and Build, Systems Integration, and Computer Support and Operations communities that integration and effective re-configuration of large, integrated systems could not be achieved within the constraints of existing processes, tools, and culture. Technical and process problems were evident in billing systems, multi-layered client/server applications, network management systems, and computer-critical exchange systems. If the problems associated with rapid growth and complexity of new technologies and large scale business solutions were not addressed, and quickly, future projects would soon run into costly overruns, suffer quality degradation, and would consume disproportionate resources on maintenance and support. Early in 1996, department managers within Design and Build and Systems Integration departments instigated an investigation tasked to provide a clear definition of the problem and to propose a solution. Results of the investigation, led by a senior integration manager, show there were seven major problem areas.

- Scale, complexity, variety, and global solutions

It's important to recognise that these factors are concerned with the systems themselves, and their interconnection to form even larger systems, rather than the files of source code and other such, lower level, components. The management and the control of these are understood and have been practised within the IT industry for many years.

- Visibility of the big picture in terms of solutions, systems, capabilities, and their interrelationships

Often, systems built in isolation to satisfy one set of demands are subsequently, by design or evolution, combined with others to form an integration hierarchy in which many different sets of demands are satisfied. Enhancement, reconfiguration, and integration happen repeatedly as new products and services are added to the company's portfolio and old ones are modified or withdrawn. Systems, once isolated, become components that must interact and share their features or capabilities with others. However, the contribution from each component is not always visible, or adequately understood, in the enlarged production, operational and managerial environment.

- Impact of change

Where suitable 'big-picture' visibility has been achieved, it is difficult to maintain by virtue of the dynamic nature of the individual, underlying components and lack of supporting processes and tools. Source files and other documents are constantly being checked out, modified, and replaced by newer versions according to good configuration management practice as demands for increased functionality are satisfied. However, much of this activity is carried out in the local production and/or operational environments. The consequences, or side effects, on other directly or indirectly connected systems are not considered until a significant, sometimes catastrophic, failure occurs. This reactive process is costly in downtime and diagnostic terms.

- Management of reuse

If BT is to remain competitive, it must provide a continuous stream of innovative, new products and services and also respond rapidly to incoming market and regulatory demands. No matter how proficient BT's systems engineering is, these goals cannot be achieved by building new systems from scratch each time. More efficient use must be made of the systems already acquired, understood, and supported. This does not refer to code modules or existing production or operational processes; the emphasis is on re-configuration and integration at the system or service platform levels.

- Definition of whole-life process deliverables

Large systems need a long time to mature and pass through many transformations, from seed of an innovative idea through production and subsequent operational use. Indeed, most of the life of an operational system may be in the post-production, maintenance phase [10]. Traditionally, expenditure during this phase was perceived as cost-of-failure because of the, mainly, diagnosis and repair nature of maintenance activities. New versions of a system were seen as new products with their own production and maintenance phases. This is no longer the case. The introduction of more efficient software engineering methods and tools and better-trained people means there are fewer code-level failures. Change during the maintenance phase is more likely to be as a result of re-configuration to produce new facilities from current system capabilities than it is to result from repair. In this scenario, detailed knowledge of a system's evolutionary history is more essential than ever. This can only be achieved within a framework of whole-life process records, process deliverables, the content and timeliness of which is clearly defined and rigorously applied.

- Labour intensive, slow audit/reporting process

There have been many quality management system audits since the quest for BS5750 registration in the early 1980s, but that is not the target here. Success or failure of the large-scale integration project depends, among other things, on an up-to-date knowledge of progress, quality, and the status of deliverables. This is not unlike other, less-integrated, system projects except that the large integration is less tangible and involves decisions that are wider reaching, and often have greater commercial impact, than their single-system counterparts. Also, there is little or no proprietary support for tracking and decision making at this level. As a result, analysis and reporting of progress and status is mainly manual, time consuming, and costly. Costs saved by providing on-line, up-to-date information could be reinvested in better quality deliveries.

- Strategic planning and evolution are disjoint from production and deployment

Within large organisations it is not uncommon to find several, major but different, sub-organisations with their own specialist skills and disciplines. This type of structure is needed to define and implement large, computer-critical systems. In BT the sub-organisations are identified with systems evolution and design, production, and deployment. The size, culture, and professional differences inevitably produce, over many years, an unseen barrier to smooth progress of system solutions. These barriers are inefficient, cause delays, and are counter-effective where the order of the day is fast, visible impact, delivery of major systems.

It is clear from the above problem statement, that any solution must focus on configuration management activities of identification, control, status accounting, and audit. However, these activities must be targeted at the higher levels of the systems integration hierarchy and interface to the lower levels where it is necessary to maintain consistency, accuracy, and, most of all, confidence.

7. Finding a solution

At the beginning April 1996 the investigation had been completed and a report containing problem definition and a proposed solution to the large-scale integration problem, was ready for presentation. During the ensuing few weeks the following outline objectives were agreed and the Integrated Configuration Management (ICM) project (to design and implement an N&S CM environment [11]) had been established. The solution must:

- be whole-life (strategy to withdrawal),
- provide early success on critical paths,

- be accepted by the whole of N&S,
- be derived mainly by harmonisation of existing methods and tools.

The ICM project was to become the primary component of a BT enterprise CM strategy [2]

7.1 Getting the message across

Given the emotive nature of control, it was decided during the initial stages of the ICM project that potential users of the enterprise CM system should accept the system for its value, rather than have it forced upon them by senior management mandate. There were two major initiatives established to support this strategy:

- a steering group composed of senior N&S managers was formed at the beginning of the project, meeting quarterly, providing authority, and demonstrating commitment to enterprise CM — it is chaired by the Systems Engineering operations manager,
- during the first few months of the project, a series of formal and informal presentation and feedback sessions were undertaken, being held at all major UK locations, highlighting the need for ECM and gathering useful user-perception information.

The main communications exercise was completed during the first quarter of 1996 but, since then, there have been several shorter sessions at BT Laboratories. Ongoing communication is seen as an essential element of ECM implementation strategy.

7.2 Design perspective

An enterprise CM system must cater for many different viewpoints and deliverables arising from many different technical and managerial environments. It must also provide decision support and interfaces to other data repositories.

Figure 9 provides a logical view of these requirements as they are implemented in the BT enterprise CM system.

The view is split between the version control environment and the configuration management environment (CME). The thick black line provides the logical boundary between the two. Below this line there exists the usual project and financial management, production, and deployment data repositories. A lot of the information is managed and controlled within proprietary configuration management systems. However, there are few information channels between individual version control environments and little opportunity to assess the impact of

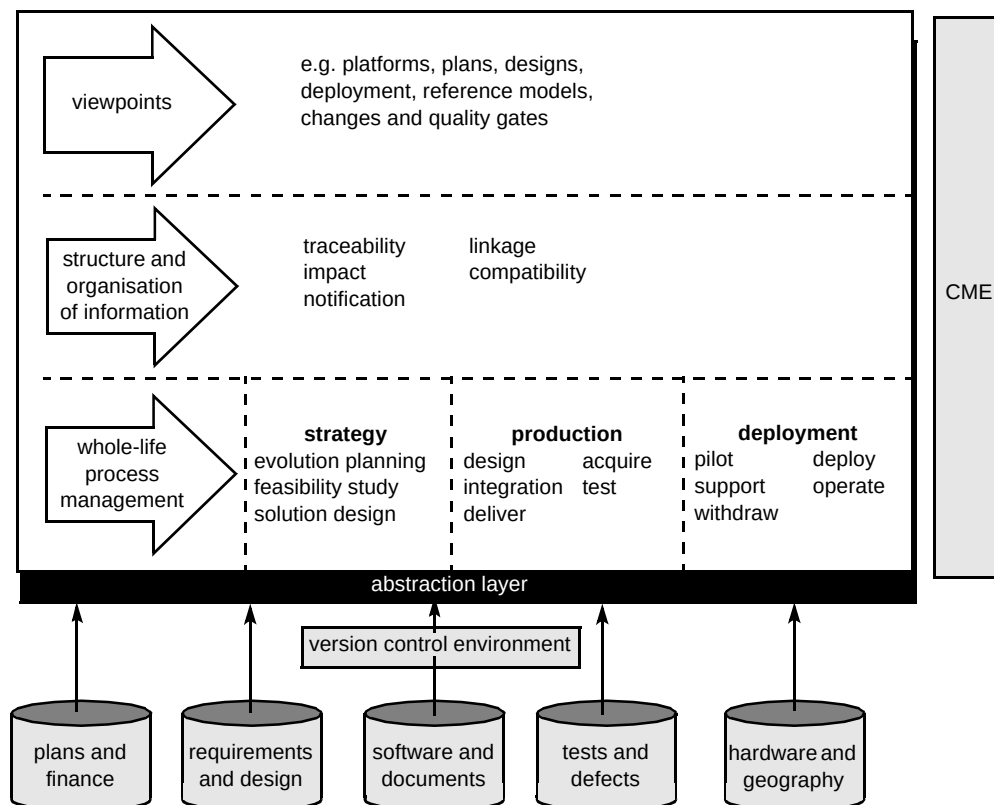


Fig 9 A logical view of enterprise CM.

change between one environment and the next. For the same reason, there is little interaction across the strategy/production and production/deployment boundaries as integrated systems progress along their evolution paths. Mostly, the configuration items consist of physical deliverables such as plans, specifications, designs, reports, inventories, and floor layouts.

On the CME side of the boundary line there lies a more abstract view of configuration management and control. The CME provides flows of filtered information across lower level, version control environments and upwards through the CM hierarchy. It is not necessary to propagate **all** change information from one environment to the next, just the information that affects decisions in the other environment or life cycle stage. If this filtering mechanism did not exist the amount of change would be unmanageable.

The CME must provide much more than information channels. It must also define and automate skeletal processes and controlled deliverables that allow the enterprise CM system to behave in a consistent, standardised fashion. It must also provide for many different types of user viewpoint — from those who are interested in detailed technical information, to those who are interested in an executive overview of progress or delivery targets. The user interface must be comprehensive but intuitive, and the structure and organisation of information must be flexible and formulated according to current business logic [12—16].

The integration of version control environments and the CME provides an enterprise CM framework within which BT can manage and control, among other things, its large-scale systems integration effort and resources. Moor et al [2] describe the implementation and deployment of the CME.

8. ECM research

Enterprise configuration management, as described in the preceding sections of this paper, is a relatively novel concept within BT. Although a pilot trial has been completed and national deployment has commenced, there are several outstanding issues to be addressed if maximum benefit is to be derived from the service platform framework and the CME.

At the present time, as there are no off-the-shelf solutions to these issues, research contracts have been placed with a number of universities.

- ECM benchmarking

Where is BT enterprise CM in relation to other large telecommunications, aerospace, and automotive enterprises? What benefits can we expect? What measures should others and BT have in place as a basis for on-going performance measurement and improvement? Is there a best ECM model?

- Modelling of component capability

How can we define capabilities as distinct from functionality — particularly non-functional capability? What is an appropriate capability model? How can this be integrated with the BT enterprise CM system?

- Visualisation of data complexity

What information structures and graphical representations are needed to provide a means of displaying the overwhelming quantity and complexity of data accessible by the enterprise CM system?

- ECM integrated business logic

What models and techniques are needed to increase the decision support, impact notification, and change control capabilities of ECM?

Some of this research work has been under way for several months and already produced a useful contribution to the ECM.

A simple capability model, based on CME and the platform framework, has been already produced and a prototype system demonstrated. A three-dimensional view of a large-scale integration has also been demonstrated.

There are still some difficult conceptual issues to resolve within these research topics, but an ECM research strategy and project management structure is in place to ensure incremental results are efficiently downstreamed, as they become available.

9. Conclusions

Many critical failures of today's major systems are the consequence of inadequate management and control over an integrated set of components. Also, not enough use is being made of bought-and-paid-for system capability. This paper has introduced the notion of an enterprise management and control system based on principles usually associated with the configuration management in an engineering environment. The BT ECM system is much more than configuration management and, at the same time, much more than a managerial control system.

There is still a lot of work to do before some of the novel issues are overcome and ECM becomes business-as-usual, but the potential business benefits have been quantified and are significant.

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Steve Moor graduated from the University of Sussex in 1974 with a degree in Electronic Engineering and joined BT, to take up a post in transmission planning and works. He joined systems development in 1979 to work on System X. In 1984 he joined Network Management Department (NMD). In 1992, after several years working on OMC and TNS in development and project management roles, he took up a post to establish a single point of systems design within NMD.

Two years later, when the design unit was established and consistent processes and methods to address solution design as well as system and deployment design had been adopted, he moved to Systems Integration department. He became the integration manager for INET. During this time the need for enterprise CM within BT became more critical. He has been managing the ICM project since 1995.



Jim Gunne-Braden joined BT as an apprentice in 1962. He received a BSc Hons (Eng) degree from the University of Ulster in 1980 and an MBA in private sector management from Essex University in 1993. Since he graduated in 1980, he has been instrumental in the design, development, and deployment of several major BT integration projects including the City Business Systems Integrated Trading System, Abbey National flotation, Work Manager, and the CME programme.



Keith Gleen joined BT as an apprentice in 1962. Following a period in London, he moved to the research department, where, in the early 1970s he worked on 'Telecommunications in the 1990s' — which aimed at delivering integrated and broadband services to the home. He then moved into computing and on to System X. From System X he was involved in the early years of research into packet switching and a variety of jobs involved with data communications. In 1990, he moved to the network management area where he became involved in future evolution of a set of core systems and components that

led to the evolution and establishment of the platform framework. Until December 1996, he was programme manager for the Enterprise Configuration Management programme. He is now innovation manager in Networks and Systems business development.

